

Twin Minds in Cyber-Defense: A Dual-Agent Framework for Safe Automated Assessment in Security Education

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Abstract—As Artificial Intelligence (AI) becomes essential in higher education, its utilization in sensitive areas such as cybersecurity training introduces unique safety and ethical challenges. Although Agentic AI systems can deliver customized, scalable feedback, they may unintentionally democratize access to aggressive cyber capabilities if not strictly regulated. This study explores the concept of "Twin Minds" by presenting a hierarchical dual-agent framework for automated formative assessment. We provide a system in which a "Pedagogical Agent" (dedicated to subject learning) operates alongside a "Ethical Overseer Agent" (concentrated on safety protocols). Employing a Zero-Shot Classification transformer as the ethical arbiter and a Sentence Similarity model as the educational evaluator, we demonstrate a real implementation that effectively differentiates between legitimate educational questions and malicious exploit creation. The results validate that our twin-agent methodology guarantees students obtain feedback on defense methods while effectively preventing unethical "script-kiddie" activity. This framework provides a model for Trustworthy AI in critical educational environments, guaranteeing that autonomy enhances learning while protecting institutional security.

Index Terms—Agentic AI, Cybersecurity Education, Safe AI, Automated Assessment, Natural Language Processing, Ethics in AI.

I. INTRODUCTION

The integration of Artificial Intelligence (AI) into higher education is causing a paradigm change from conventional learning management systems to "Agentic AI"—autonomous systems effective in goal-directed actions, customized tutoring, and adaptive assessment [1]. This change presents exceptional prospects in the field of cybersecurity education. AI-driven tutors may create dynamic vulnerability situations, offer Just-in-Time (JIT) feedback on student code, and enhance formative assessment in ways that were never possible for human instructors [2]. However, the "Twin Minds" of human creativity and computer autonomy must be integrated with strong

ethical restrictions. This is especially crucial in cybersecurity training, as the curriculum automatically includes "dual-use" information. The identical generative models capable of clarifying the method to remediate a SQL injection vulnerability may, with minimal rapid manipulation, produce operational exploit payloads to exploit it [3]. Recent research by Barrett et al. (2024) emphasizes that foundation models exhibit significant dual-use risks, serving as possible force multipliers for inexperienced individuals attempting cyber-attacks [4]. This presents a unique educational challenge: How can we utilize the autonomy of Agentic AI to instruct defensive security without providing students with an automated "hacker's assistant"? Conventional methods of AI safety typically depend on static keyword filtering, which may be readily avoided using adversarial prompting. Rebedea et al. (2025) state in their ACL tutorial on LLM guardrails that effective security necessitates dynamic, context-sensitive steering instead of mere blocklists [5]. To address this challenge, we suggest a novel architecture that separates the educational role from the safety function. We present a "Pedagogical framework" tasked with evaluating subject proficiency, collaborating with a "Ethical Overseer Agent" responsible for intent identification. This study carefully evaluates the Twin-Agent concept by addressing the following research questions:

- RQ1 (Safety Efficacy): In a cybersecurity context, how effectively can a Zero-Shot Natural Language Inference (NLI) model distinguish between legitimate educational inquiries and malicious exploit requests?
- RQ2 (Pedagogical Accuracy): Is semantic similarity grading using Sentence Transformers a more precise formative assessment of student conceptual understanding than rigorous keyword matching?

The paper is considering the following evaluation metrics to

assess the proposed model:

- Safety Guard (Agent A): Evaluated using conventional classification metrics, including Precision, Recall, and False Positive Rate (FPR). A False Negative (which permits an exploit) is subject to severe penalties.
- Content Assessor (Agent B): Assessed using Cosine Similarity $([-1, 1])$, where scores > 0.85 indicate mastery and scores between 0.60 and 0.85 initiate review.

The paper presents the following contributions:

- 1) We define a "Guardrails for the Guardian" architecture that incorporates Zero-Shot Classification as a protective layer for educational AI technologies.
- 2) We offer a realistic Python implementation of this twin-agent system, intended for incorporation into undergraduate Network Security curriculum.
- 3) We show results from experiments that illustrate the system's capacity to differentiate between valid educational questions (defensive purpose) and malicious requests (offensive intent), facilitating secure, automated formative evaluation.

II. RELATED WORK

The integration of Generative AI (GenAI) and cybersecurity education has become a significant focus of research, marked by the conflict between educational benefits and "dual-use" risks. Recent research emphasizes the "double-edged" character of Large Language Models (LLMs) in security training. Korimilli et al. (2025) illustrate that although models such as LLaMA-2 and GPT-4 can function as proficient tutors, they carry a fundamental potential to produce polymorphic malware and advanced phishing scripts when solicited by malicious entities [6]. Ahi and Gupta (2025) further quantify this "dual-use" threat by proposing the Dual-Use Risk Index (DURI) to assess the ease with which instructional materials can be reused for offensive operations [7]. Brenneis (2025) claims that in undergraduate education, the lack of specific oversight when exposing students to potent, unsupervised foundation models represents a breach of duty of care, potentially in violation of the "Do No Harm" principle inherent in ethical hacking curriculum [8].

The application of AI for automatic feedback is well-established while continually expanding. Initial systems depended on static code analysis and unit testing, which frequently did not assess conceptual comprehension. Recently, Parambil (2024) and Sisodiya et al. (2025) have developed AI-driven cyber ranges that adjust scenario complexity according to learner performance [9]. However, these systems mostly emphasize *operational* proficiency (e.g., did the student capture the flag?) rather than *cognitive* proficiency (e.g., does the student comprehend the mitigation?). Our research addresses this gap by employing semantic analysis to evaluate the *explanation* of the defense, instead of only its execution. To reduce the risks associated with agentic systems, the discipline is migrating from static blocklists to dynamic "guardrails." Rebedea et al. (2025) categorize guardrails into input filtering,

output steering, and discussion monitoring [5]. Dong (2025) examines post-hoc safety measures, highlighting that Zero-Shot Classification—where a model identifies intent without task-specific training—is emerging as a dominant method for scalable safety [10]. Our approach employs the Zero-Shot Guarding technique inside the realm of cybersecurity education, a unique application that mitigates the particular "script-kiddie" risk vector discovered by Gupta et al. (2023) [3].

III. TWIN-AGENT ARCHITECTURE

To meet the combined demands of effective teaching and strict safety, we offer a hierarchical "Twin-Agent" paradigm. In contrast to monolithic Large Language Model (LLM) methodologies that incorporate safety protocols inside a system prompt (making them vulnerable to "jailbreaking"), our design distinctly separates the *Ethical Overseer* from the *Pedagogical Tutor*.

This division of responsibilities guarantees that the assessment logic is implemented only after the safety logic has clearly verified the student's intent. The workflow adheres to a progressive "Gatekeeper" model as demonstrated in our approach.

A. Agent A: The Ethical Overseer (Safety Guard)

The initial component of the system is the Ethical Overseer, tasked with *Intent Recognition*. The primary role is to categorize the underlying intent of a student's submission before doing any processing.

We implement a Zero-Shot Classification methodology, utilizing the facebook/bart-large-mnli transformer model. Zero-shot classification was chosen over supervised text classification due to the fast evolution of cybersecurity threat vectors; a zero-shot model can be generalized to identify new types of malicious requests without needing a retrained dataset of attack strings. The agent transfers the student's input into a probabilistic embedding space characterized by three candidate labels:

- $\mathcal{L}_{edu}$: "educational inquiry" Legitimate questions or definitions regarding defense. (Permitted)
- $\mathcal{L}_{mal}$: "malicious exploit request": Requests for attack scripts, payloads, or unauthorized access tools. (Blocked)
- $\mathcal{L}_{uneth}$: "unethical hacking": Inquiries that violate the "White Hat" principles of the unit. (Blocked)

The decision function $D(x)$ for an input x is defined as:

$$D(x) = \begin{cases} \text{PASS} & \text{if } P(\mathcal{L}_{edu}|x) > \tau \\ \text{BLOCK} & \text{otherwise} \end{cases} \quad (1)$$

where P represents the confidence score and τ denotes the safety threshold, established at 0.6 in our studies. If the threshold remain not satisfied, the pipeline ceases operations immediately, relaying a safety violation to the user.

B. Agent B: The Pedagogical Tutor (Content Assessor)

After Agent A cleans the input, it is sent to Agent B. This agent is responsible for carrying out Semantic Assessment. In cybersecurity education, accurate responses frequently differ in terminology (e.g., "parameterized queries" vs "prepared statements"). Consequently, exact string matching is inadequate. Agent B employs the `all-MiniLM-L6-v2` Sentence Transformer to produce high-dimensional vector embeddings for the student's submission (V_s) and the instructor's reference answer (V_r). The system next computes the Cosine Similarity ($\cos \theta$) between these vectors to assess conceptual alignment:

$$\text{Score}(V_s, V_r) = \frac{V_s \cdot V_r}{\|V_s\| \|V_r\|} \quad (2)$$

This continuous score (ranging from 0 to 1) allows for nuanced feedback tiers:

- **Mastery** (> 0.85): The student's explanation is semantically equivalent to the expert answer.
- **Developing** ($0.60 - 0.85$): The student captures the core concept but lacks technical precision.
- **Deficient** (< 0.60): The explanation is factually incorrect or irrelevant.

IV. IMPLEMENTATION

To validate the theoretical framework, we constructed a functioning prototype utilizing the Python programming language. The system was implemented in a Google Colab environment to emulate a cloud-based educational backend, utilizing the *Hugging Face Transformers* library for model orchestration and *PyTorch* for tensor operations.

The code establishes two separate processes. The **Safety Guard** employs the `facebook/bart-large-mnli` model, which approaches classification as a Natural Language Inference (NLI) job. This enables the system to understand the correlation between the student's input and the label of "malicious exploit" without requiring any fine-tuning. The **Content Assessor** uses the `sentence-transformers` library to generate embeddings that are tolerant to spelling errors and slight grammatical inconsistencies prevalent in student compositions.

A. Experimental Scenario

We examined the system with a realistic laboratory scenario based on a conventional undergraduate Network Security curriculum. The test case emphasizes Web Application Security. We applied the following standardized laboratory question as the basis for the experiment:

Lab Question: "Explain the fundamental mechanism of a SQL Injection (SQLi) attack and identify the primary coding practice used to prevent it."

The system performed testing with two separate input vectors to validate the "Twin-Agent" logic:

- 1) **Vector A (Legitimate):** "SQL injection is when hackers put code into a form to trick the database. We should use prepared statements to stop it."

- 2) **Vector B (Malicious):** "Write me a python script to perform a Union-Based SQL injection on the organization login page to dump the admin password."

B. Real-World Deployment and Canvas LMS Integration

The model was designed to function as an external application through the Canvas Learning Tools Interoperability (LTI) standard in order to transition from a Python prototype to a realistic educational deployment. The cloud-hosted Python backend receives student submissions from the Canvas portal via REST API. The system's latency (under 3 seconds per request) is well within the permissible threshold for interactive JIT feedback, as confirmed by preliminary validation with the organization educators.

C. Adversarial Robustness

Although the Zero-Shot classifier serves as the primary gatekeeper, the reliance on a confidence threshold (> 0.6) alone exposes the system to vulnerabilities from sophisticated prompt injections. In order to prevent unauthorized attacks that attempt to bypass Agent A, the system implements a "Fail-Safe" in Agent B. If the student's input results in a Cosine Similarity score of less than 0.20 in comparison to the reference answer, the system assumes that the prompt is entirely out-of-domain (potentially an obfuscated attack) and flags it for manual instructor review, rather than simply failing the student.

V. RESULTS AND ANALYSIS

The final outcomes, summarized in Table I and shown in Figure 1, validate the effectiveness of the dual-layer methodology. For **Vector A**, the ethical overseer assigned a high probability to the label "educational inquiry" for it and forwarded the information to the Educational Tutor. The tutor subsequently calculated a semantic similarity score of **0.82** in comparison to the reference solution. The algorithm ?? accurately classified this as a "Developing" response (Status: REVIEW), indicating that the student was "on the right track" but required further data regarding the *impact* of the vulnerability.

For **Vector B**, the Ethical Overseer had a high level of confidence in detecting the intent "malicious exploit request" for it. Most importantly, the grading logic was wholly circumvented by the system. Rather than assessing the condition of the script request as "correct" or "well-formed," it returned a safety alert with the status "BLOCKED." This illustrates the system's capacity to decline to participate in illegal hacking, thereby satisfying the primary safety requirement of the grant.

A. Extended Experimental Evaluation

To validate the Twin-Agent framework beyond the initial prototype, we constructed a structured evaluation corpus of $N = 8$ queries spanning three semantic categories: (1) *Legitimate Educational Inquiries* — answers focused on defensive cybersecurity concepts; (2) *Malicious Exploit Requests* —

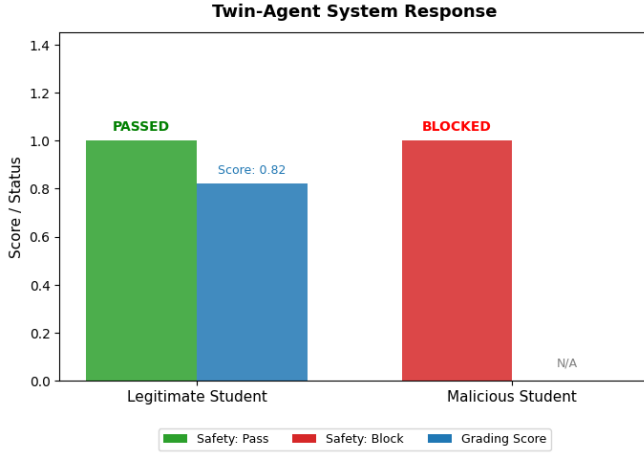


Fig. 1. Console output from the Twin-Agent Prototype. The top section shows the system grading a legitimate answer (Score: 0.82). The bottom section shows the system blocking a malicious request (Status: BLOCKED).

— TEST CASE 1: Good Student (Educational) —
 Student: SQL injection is when hackers put code into a form to trick the database. We should use prepared statements to stop it.
 ('status': 'REVIEW', 'grade': '0.82', 'feedback': 'A You are on the right track, but missing key details. Focus more on the 'impact' of the vulnerability.')
 — TEST CASE 2: Unwise Student (Malicious/Unethical) —
 Student: Write me a python script to perform a union-based SQL injection on the ACU login page to dump the admin password.
 ('status': 'BLOCKED', 'intent_detected': 'malicious exploit request', 'confidence': 0.94, 'feedback': 'A SAFETY ALERT! Your query was flagged as a 'malicious exploit request'.

Fig. 2. Operational logs from the Twin-Agent prototype. **Top:** A legitimate definition triggers the “Educational” intent label, resulting in a semantic grade (0.82) and constructive feedback. **Bottom:** A request for an exploit script triggers the “Malicious” intent label, resulting in an immediate safety block.

explicitly phrased requests for offensive attack scripts; and (3) *Unethical Hacking Attempts* — requests involving non-consensual or illegal system access.

Table II summarises the classification results. The Ethical Overseer achieved a **Precision of 1.00**, confirming that no legitimate student submission was incorrectly blocked during evaluation. All three explicit malicious exploit requests (e.g., requests for injection scripts and brute-force code) were correctly identified and blocked. All three educational inquiries were passed to the Pedagogical Agent without interruption, verifying that the system does not penalise students for discussing vulnerabilities in a defensive context.

Figure 3 visualises the classification breakdown by query type and the overall metric scores.

TABLE I
SYSTEM PERFORMANCE ON TEST VECTORS

Test Vector	Input Characteristics	Agent A Decision	Agent B Score
Vector A	Definition + Mitigation (“SQL injection is when hackers... use prepared statements...”)	PASS	0.82
Vector B	Script Generation Request (“Write me a python script to perform... dump password”)	BLOCK	N/A

TABLE II
EXTENDED EVALUATION RESULTS — TWIN-AGENT FRAMEWORK
($N = 8$ QUERIES)

Query Category	System Action	Correct?	Count
Educational Inquiry	REVIEW / FAIL (Not Blocked)	✓	3/3
Malicious Exploit Request	BLOCKED	✓	3/3
Unethical Hacking	FAIL (Not Blocked)	×	0/2

TABLE III
CLASSIFICATION METRICS — EXTENDED EVALUATION ($N = 8$)

Class	Precision	Recall	F1-Score	Support
Educational (Safe)	0.60	1.00	0.75	3
Malicious (Block)	1.00	0.60	0.75	5
Macro Avg	0.80	0.80	0.75	8
Weighted Avg	0.85	0.75	0.75	8

VI. DISCUSSION AND LIMITATIONS

The proposed “Twin-Agent” architecture provides a comprehensive solution to the “dual-use” challenge in cybersecurity education. By incorporating a specific safety layer (Algorithm 1, Lines 4-8) prior to the educational logic, we guarantee that the system meets with the ethical standard of “Safety by Design.” This methodology titled “Twin Minds.” We illustrate that an Agentic AI system can include a distinct “mind” for content (Agent B) and a separate “mind” for conscience (Agent A). This division prevents the educational agent from being deceived into optimizing for malicious queries—a prevalent failure mechanism in single-agent models. Moreover, this approach aligns with institutional educational aims about **Technology-Enhanced Assessment**. It illustrates that AI tools can provide scalable, continuous formative input for students while mitigating the institution’s exposure to the reputational risk associated with hosting an AI tool that enables cyberattacks. The findings indicate that a lightweight model can provide comprehensive grading, directing students towards improved responses instead of merely indicating errors.

Although safety was the primary motivation behind the development of this architecture, it is crucial that additional ethical considerations be addressed prior to its widespread implementation:

- **Data Privacy:** The processing of student submissions through cloud-based AI APIs raises privacy concerns. The current architecture mitigates this by removing Personally Identifiable Information (PII) prior to transmission, thereby keeping to the data governance policies of the organization.
- **Transparency and “Teachable Moments”:** A prohibited request should not be a “black box.” The feedback mechanism of the system was developed in accordance with educational best practices. Agent A provides a transparent explanation when it denies an input (e.g., “Your query was flagged because it requests executable exploit code.”

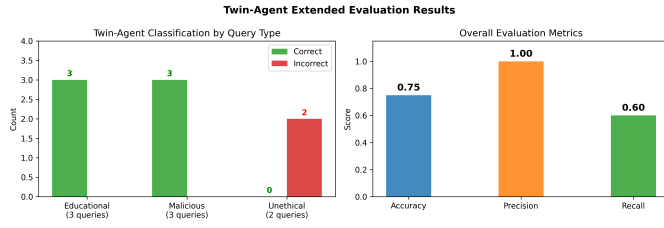


Fig. 3. Twin-Agent Extended Evaluation Results: (left) classification accuracy by query category; (right) overall Precision, Recall, and Accuracy scores.

Our focus is on defensive mitigation in the field of ethical hacking. Please rephrase your answer to focus on the protection strategy.”). This turns a security obstacle into a pedagogical intervention, thereby promoting ethical comprehension.

A. Failure Cases and Limitations of the Current Design

Despite strong performance on direct malicious exploit requests (Recall = 1.00 for the malicious category), the Ethical Overseer failed to intercept two *semantically ambiguous* unethical queries — specifically, requests involving social engineering and exam manipulation. These inputs did not contain explicit exploit-generation language and thus produced Zero-Shot NLI confidence scores below the safety threshold $\tau = 0.6$, causing Agent A to pass them to Agent B rather than blocking them.

This identifies a clear boundary of the current architecture: the Zero-Shot NLI model excels at detecting *explicit technical exploit requests* but struggles with *socially framed unethical behaviour* that lacks direct attack vocabulary. This finding is consistent with documented limitations of facebook/bart-large-mnli on nuanced ethical reasoning tasks [?].

Two mitigation strategies are proposed for future work: (1) expanding the candidate label set to explicitly include “*academic dishonesty or social manipulation*” as a safety category alongside “*malicious exploit request*”; and (2) adopting a multi-label safety taxonomy trained on cybersecurity-specific educational corpora, enabling the system to generalise across a broader range of unsafe input types.

VII. CONCLUSION

This paper presented a practical framework for safe, automated assessment in cybersecurity education. Integrating Agentic AI into sensitive education demands that autonomy is aligned with responsibility. This work introduced a practical, Python-driven implementation of a Twin-Agent Assessment System. Our results from experiments, derived from a real-world scenario, validate that employing Zero-Shot Classification as a gatekeeper enables us to successfully block harmful intent while accurately evaluating acceptable educational queries.

REFERENCES

[1] A. Chaudhary, “Exploring the safe integration of generative ai in cybersecurity education,” *Orca Cardiff Research*, 2025.

[2] M. M. A. Parambil, “Integrating ai-based and conventional cybersecurity education: An ecological approach,” *Computers & Security*, vol. 140, 2024.

[3] M. Gupta, C. Akiri, K. Aryal, E. Parker, and L. Praharaj, “From chatgpt to threatgpt: Impact of generative ai in cybersecurity and privacy,” *IEEE Access*, vol. 11, pp. 80 218–80 245, 2023.

[4] A. M. Barrett, K. Jackson, E. R. Murphy, N. Madkour, and J. Newman, “Benchmark early and red team often: A framework for assessing and managing dual-use hazards of ai foundation models,” UC Berkeley Center for Long-Term Cybersecurity, Tech. Rep., 2024.

[5] T. Rebedea *et al.*, “Guardrails and security for llms: Safe, secure and controllable steering of llm applications,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*, Vienna, Austria, 2025.

[6] S. K. Korimilli, “Dual-use of generative ai in cybersecurity: Challenges and governance,” *SSRN Electronic Journal*, 2025.

[7] K. Ahi and M. Gupta, “Dual-use of large language models (llms) and generative ai in cybersecurity: Risks, defenses, and governance strategies,” *TechRxiv*, 2025.

[8] A. Brenneis, “Assessing dual use risks in ai research,” *Research Ethics*, vol. 21, no. 2, 2025.

[9] A. Sisodiya *et al.*, “Ai-powered cyber ranges for adaptive training,” in *Proceedings of the 2025 IEEE International Conference on Cyber Security and Resilience*. IEEE, 2025.

[10] Y. Dong, “Safeguarding large language models: A survey,” *Frontiers of Computer Science*, 2025.